

DIRE DAWA UNIVERSITY

Institute of Technology · School of Computing · Department of Software Engineering

INTERNSHIP PRESENTATION

Membership Management System Modernisation at Afronex Tech Company

Student: Nebiyu Tsegaye | **ID:** 1501332

Programme: BSc in Software Engineering

Internship Organization: Afronex Tech Company, Dire Dawa

Internship Period: 10/06/2018 E.C – 10/10/2018 E.C

Industry Supervisor: Mr. Tsegaw

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Following the presentation: live project demo

1. Introduction

- **Afronex Tech Company** — Dire Dawa, Ethiopia; technology solutions in web dev, system dev, and AI
- **Internship period:** 10/06/2018 E.C – 10/10/2018 E.C (4 months)
- **Main project:** Modernisation of a membership management system — rewrite + new features
- **My role:** Full-stack developer; Team 6 (Payments ★) + Team 2 (Super Admin ★); communication lead; AI engineer
- **Other projects:** AI Care (Django), Accounting (Odoo), personal project Temariware

2. Problem Statement

- **Outdated tech stack:** Existing system had accumulated technical debt limiting extensibility
- **No digital ID:** Member registration relied on manual ID collection — no Fayda support
- **Limited payments:** No Telebirr integration; payments collected manually with no audit trail
- **Rigid schema:** No custom attributes — every organisation had the same fixed member fields
- **Weak verification:** No OTP during registration; no screenshot-based payment verification

3. Project Objective

General: Modernise the membership management system with modern frameworks and six client-requested features, across seven cross-functional teams.

Specific Objectives

- Integrate **Fayda ID** for member registration
- Implement **custom attributes** for org-specific member fields
- Integrate **Telebirr payment** for fees and renewals
- Build **manual payment confirmation** + invoice workspace
- Add **OTP confirmation** for user registration
- Implement **OCR screenshot verification** for low-friction payment

4. Scope and Limitation

IN SCOPE

- Rewriting system with React + TypeScript
- Fayda ID, custom attributes, Telebirr, OTP
- OCR-based payment verification
- Manual payment confirmation workspace
- Invoice and receipt generation
- Super admin dashboard (Team 2)
- Seven-team structure with shared contracts

LIMITATIONS

- Confidentiality: client name/URLs undisclosed
- OCR accuracy depends on screenshot quality
- Manual payment review introduces latency
- Custom attributes use flexible key-value (not typed)
- No automated Telebirr verification in v1
- Internship period limited to 4 months

5. Specification

TECHNICAL STACK

- **Main project:** React + TypeScript (company-specified)
- **Database:** PostgreSQL via Prisma ORM
- **AI Care:** Django (preference-based)
- **Accounting:** Odoo (preference-based)
- **Personal:** Next.js, Tailwind, shadcn/ui
- **AI tools:** Agentic AI for scaffolding
- **OCR:** Tesseract / EasyOCR
- **Note:** No fixed tech stack obligation

TEAM ALLOCATION (7 TEAMS)

Team	Role
1	Platform & Security
2 ★	Super Admin & Governance (mine)
3	Org Admin & Member Mgmt
4	Event Management
5	Blog & Announcements
6 ★	Payments & Subscriptions (mine)
7	Integration QA

6. Project Feasibility

Dimension	Analysis	Verdict
Technical	React, TypeScript, Django, Odoo — mature, well-documented, open-source	Feasible
Economic	All tools free/open-source. Company provided infrastructure at no cost to interns	Feasible
Operational	Aligns with company workflow. Interns integrated into active teams	Feasible
Schedule	4 months, 7 teams in dependency order. Tight but achievable	Feasible

Main risk: Cross-team integration — mitigated by Team 1 locking contracts first

7. Methodology

- **Approach:** Agile — iterative development with adaptive planning and continuous feedback
- **Why Agile?** Client requirements evolved through feedback on early iterations
- **Sprint structure:** Each team owned features, delivered within agreed timeline
- **Team allocation:** Seven teams with strict dependency order
- **Workflow:** Plan → Checklist → Implement → PR Review → Integration (Team 7) → Acceptance
- **Communication:** Weekly meetings, Telegram, GitHub PRs, Zoom, in-person

8. Result and Outcomes — Features Delivered

TEAM 6: PAYMENTS & SUBSCRIPTION

- Manual payment confirmation workflow (full-stack)
- Invoice and receipt workspace
- Telebirr payment integration
- OCR-based screenshot payment verification
- Payment status normalization (audit trail)

TEAM 2: SUPER ADMIN & ORG GOVERNANCE

- Super admin dashboard interfaces
- Organization management & status actions
- Org admin oversight views
- Platform-level control features

All features passed Team 7 integration testing. Code is being tested for production deployment.

8b. Other Contributions & Personal Outcomes

OTHER CONTRIBUTIONS

- AI Care system (Django) — data flow + AI features
- Accounting & inventory subsystem (Odoo)
- Recommended features adopted by the company
- **Temariware** — personal project, live in production

PERSONAL OUTCOMES

- Shift to business-aware mindset
- Agile multi-team experience (Teams 2 + 6)
- Communication & leadership as interns' lead
- Cursor Hackathon certificate (Haramaya Univ.)

9 features · 2 teams · 1 live project

9. Conclusion and Recommendation

CONCLUSION

- Modernised the system with 6 new features
- Team 6 + Team 2 deliverables passed integration
- Key learning: business-aware mindset shift
- Temariware (personal project) now live

RECOMMENDATIONS

- Integrate Chapa API for automated verification
- Formalise onboarding for future interns
- Continue exploring agentic AI with oversight
- Future: specialise in AI engineering

Next: Live project demo

Thank You

Questions & Answers

Nebiyu Tsegaye

ID: 1501332

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Project Demo: Membership Management System + Temariware